Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Technology, Pune-37



Department of Computer Engineering

Class and Division	CS-C
Batch Number	03
GR-No	1251070278
Roll Number	2070304
Student Name	Atharva Bhausaheb Bodke
Assignment Number	09
Assignment Title	**SQL using any suitable programming language.**
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Assignment Number: 09

Assignment Title: (SQL) using any suitable programming language.

#CODE:PYTHON

```
import mysql.connector

# Establish Database Connection
con = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Rohit@",
    database="StudentDB"
)

if con.is_connected():
    print("Connected to MySQL Database Successfully!")

cursor = con.cursor()

while True:

    print("\n===== STUDENT MANAGEMENT SYSTEM =====")
    print("1. Add Student")
    print("2. Display All Students")
    print("3. Search Student")
    print("4. Update Student Marks")
    print("5. Delete Student")
    print("6. Exit")

    choice = int(input("Enter your choice: "))

    # 1. Add Student
    if choice == 1:

        roll = int(input("Enter Roll Number: "))
        name = input("Enter Student Name: ")
```

```
    dept = input("Enter Department: ")
    city = input("Enter City: ")
    marks = float(input("Enter Marks: "))

    query = """
    INSERT INTO Students
    VALUES (%s, %s, %s, %s, %s)
    """

    values = (roll, name, dept, city, marks)

    cursor.execute(query, values)
    con.commit()

    print("Student added successfully!")

# 2. Display All Students
elif choice == 2:

    cursor.execute("SELECT * FROM Students")

    records = cursor.fetchall()

    if records:
        print("\nRoll No | Name | Department | City | Marks")

        for row in records:
            print(row)
    else:
        print("No student records found!")

# 3. Search Student
elif choice == 3:

    roll = int(input("Enter Roll Number to search: "))

    query = "SELECT * FROM Students WHERE roll_no = %s"

    cursor.execute(query, (roll,))

    record = cursor.fetchone()

    if record:
        print("Student found:", record)
```

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    else:
        print("Student not found!")

# 4. Update Student Marks
elif choice == 4:

    roll = int(input("Enter Roll Number: "))
    marks = float(input("Enter New Marks: "))

    query = """
    UPDATE Students
    SET marks = %s
    WHERE roll_no = %s
    """

    cursor.execute(query, (marks, roll))
    con.commit()

    if cursor.rowcount > 0:
        print("Student marks updated successfully!")
    else:
        print("Student not found!")

# 5. Delete Student
elif choice == 5:

    roll = int(input("Enter Roll Number to delete: "))

    query = "DELETE FROM Students WHERE roll_no = %s"

    cursor.execute(query, (roll,))
    con.commit()

    if cursor.rowcount > 0:
        print("Student deleted successfully!")
    else:
        print("Student not found!")

# 6. Exit
elif choice == 6:

    print("Application closed.")
    break
```

```
    else:
        print("Invalid choice! Try again.")

# Close Database Connection
cursor.close()
con.close()

print("Database connection closed.")



#CODE : database.sql


-- Create Database
CREATE DATABASE StudentDB;

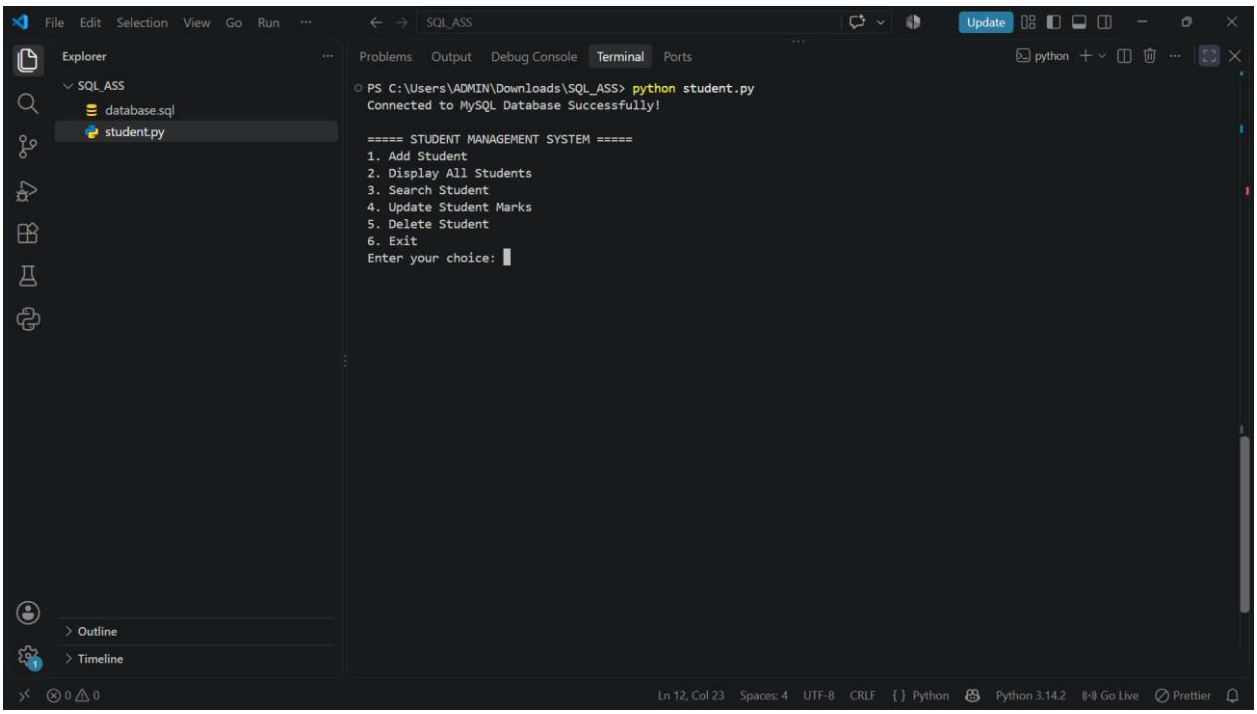
-- Select Database
USE StudentDB;

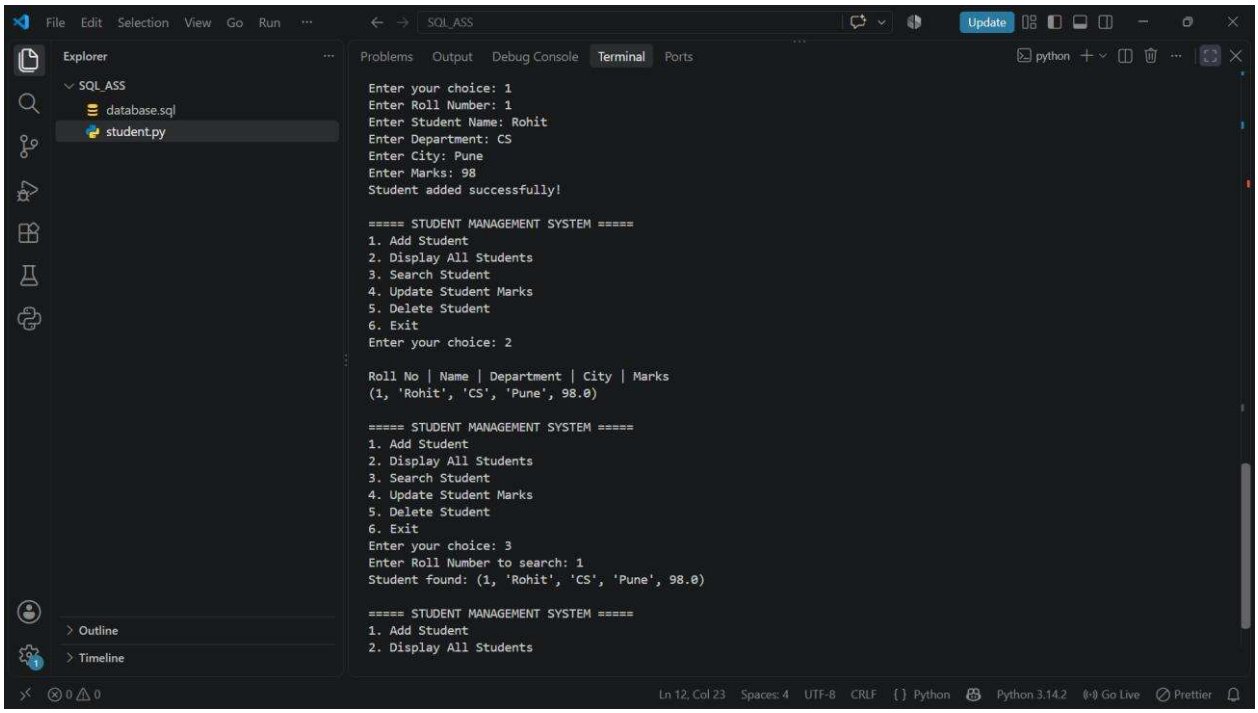
-- Create Students Table
CREATE TABLE Students (
    roll_no INT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    department VARCHAR(50),
    city VARCHAR(30),
    marks FLOAT
);

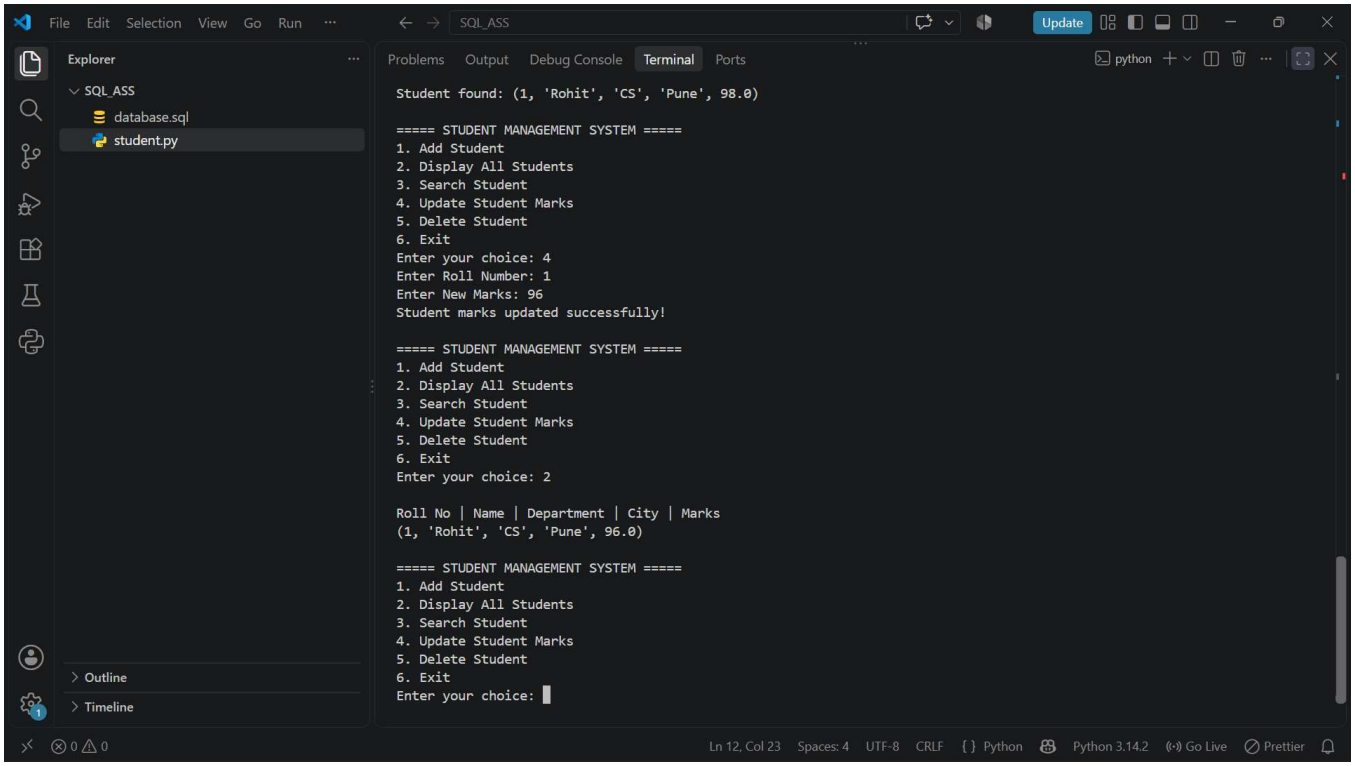
-- Display Table Structure
DESC Students;

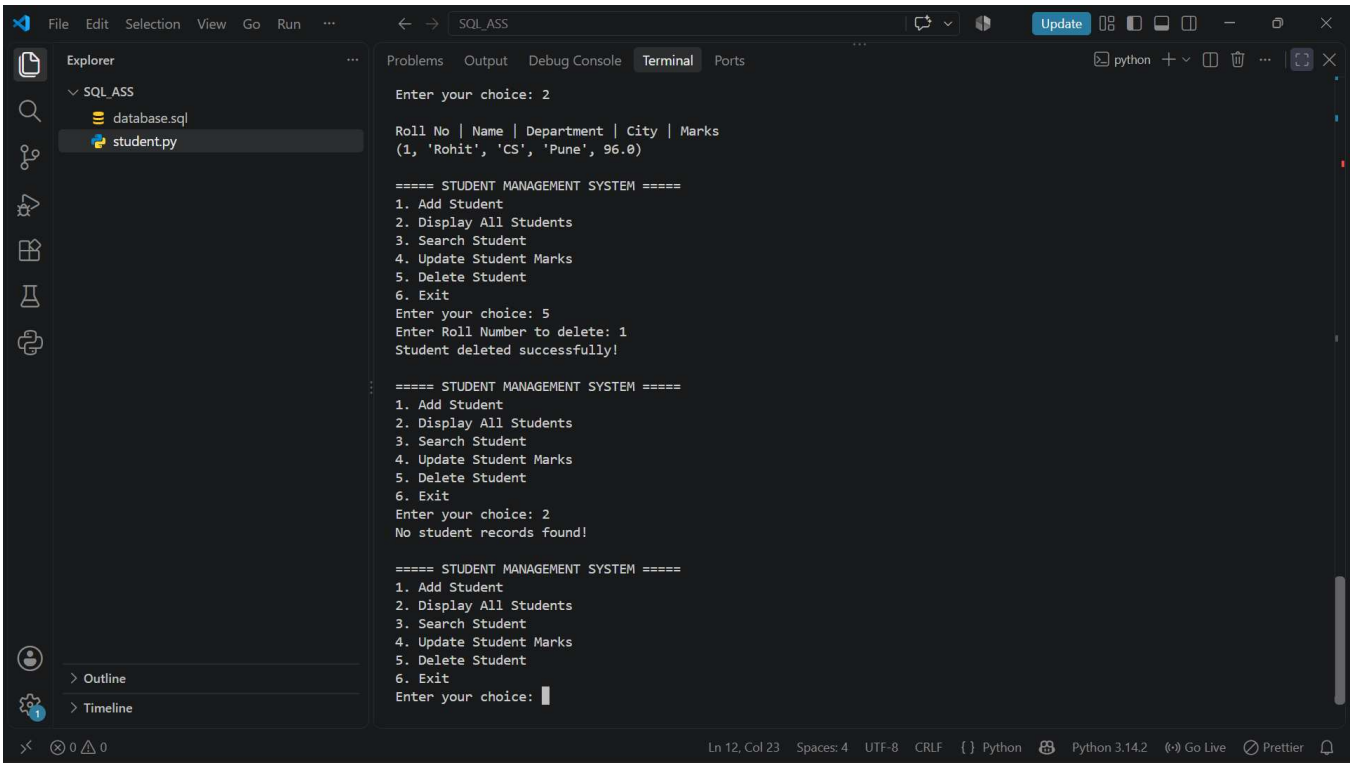
-- Display All Records
SELECT * FROM Students;
```

OUTPUT:


File Edit Selection View Go Run
SQL_ASS
Update
Explorer
SQL_ASS
database.sql
student.py
Problems Output Debug Console Terminal Ports
python
PS C:\Users\ADMIN\Downloads\SQL_ASS> python student.py
Connected to MySQL Database Successfully!
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice:
Outline
Timeline
Ln 12, Col 23 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.2 Go Live Prettier


File Edit Selection View Go Run
SQL_ASS
Update
Explorer
SQL_ASS
database.sql
student.py
Problems Output Debug Console Terminal Ports
python
Enter your choice: 1
Enter Roll Number: 1
Enter Student Name: Rohit
Enter Department: CS
Enter City: Pune
Enter Marks: 98
Student added successfully!
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice: 2
Roll No | Name | Department | City | Marks
(1, 'Rohit', 'CS', 'Pune', 98.0)
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice: 3
Enter Roll Number to search: 1
Student found: (1, 'Rohit', 'CS', 'Pune', 98.0)
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
Outline
Timeline
Ln 12, Col 23 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.2 Go Live Prettier


File Edit Selection View Go Run
SQL_ASS
Update
Explorer
SQL_ASS
database.sql
student.py
Problems Output Debug Console Terminal Ports
python
Student found: (1, 'Rohit', 'CS', 'Pune', 98.0)
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice: 4
Enter Roll Number: 1
Enter New Marks: 96
Student marks updated successfully!
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice: 2
Roll No | Name | Department | City | Marks
(1, 'Rohit', 'CS', 'Pune', 96.0)
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice:
Outline
Timeline
Ln 12, Col 23 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.2 Go Live Prettier


File Edit Selection View Go Run
SQL_ASS
Update
Explorer
SQL_ASS
database.sql
student.py
Problems Output Debug Console Terminal Ports
python
Enter your choice: 2
Roll No | Name | Department | City | Marks
(1, 'Rohit', 'CS', 'Pune', 96.0)
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice: 5
Enter Roll Number to delete: 1
Student deleted successfully!
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice: 2
No student records found!
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student Marks
5. Delete Student
6. Exit
Enter your choice:
Outline
Timeline
Ln 12, Col 23 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.2 Go Live Prettier